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Human-Centered Artificial Intelligence: Designing for User Empowerment and Ethical Considerations

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***Abstract*—Human-Centered Artificial Intelligence (AI) focuses on AI systems prioritizing user empowerment and ethical considerations. We explore the importance of user-centric design principles and ethical guidelines in creating AI technologies that enhance user experiences and align with human values. It emphasizes user empowerment through personalized experiences and explainable AI, fostering trust and user agency. Ethical considerations, including fairness, transparency, accountability, and privacy protection, are addressed to ensure AI systems respect human rights and avoid biases. Effective human AI collaboration is emphasized, promoting shared decision-making and user control. By involving interdisciplinary collaboration, this research contributes to advancing human-centered AI, providing practical recommendations for designing AI systems that enhance user experiences, promote user empowerment, and adhere to ethical standards. It emphasizes the harmonious coexistence between humans and AI, enhancing well-being and autonomy and creating a future where AI technologies benefit humanity. Overall, this research highlights the significance of human-centered AI in creating a positive impact. By centering on users' needs and values, AI systems can be designed to empower individuals and enhance their experiences. Ethical considerations are crucial to ensure fairness and transparency. With effective collaboration between humans and AI, we can harness the potential of AI to create a future that aligns with human aspirations and promotes societal well-being.**

Keywords—Human-Centered, Digitalization, Artificial Intelligence, AI, Ethicality, Ethical consideration, Empowerment, Transparency, Technology transfer, Society 2.0

I. INTRODUCTION

Artificial Intelligence (AI) has emerged as one of the digitalization [32][66] related to a transformative technology, as part of technology development in goods to services society changes [67] with the inherent potential to revolutionize our daily lives. From virtual assistants [31], digital design [33] and edition support tools [28][36], recommendation systems to autonomous vehicles [5] and healthcare diagnostics [19], AI is increasingly permeating diverse domains. However, as AI becomes more ubiquitous, it is essential to focus on proper enablers [16] and prioritize the human experience and ensure that AI systems are designed with a human-centered approach [1][2], with proper control considerations [23] and keep systems safe against unauthorized access and edits [26]. Human-Centered Artificial Intelligence, often referred as HCAI, is a paradigm that emphasizes the central role of humans in the design, development, and deployment of AI systems. It recognizes that AI should be a tool to empower individuals, enhance their capabilities, and improve their overall well-being. By placing the user at the forefront, HCAI aims to bridge the gap between machines and humanity, creating a harmonious and beneficial coexistence. Primary goal of HCAI is to design AI systems for unique preferences, needs and individual values. User-centric design principles are crucial in this process, as they ensure that AI technologies align with the expectations and aspirations of users. [3][4]By adopting a user-centric approach, AI systems can deliver personalized experiences, seamless interactions, and intuitive interfaces that enhance user satisfaction and acceptance. One of the fundamental aspects of HCAI is user empowerment. Empowering users involves granting them control, agency, and understanding of AI systems' functioning. Explainable AI, an area of research within HCAI, focuses on making AI algorithms and decision-making processes transparent and understandable to users. This not only fosters trust but also enables users to make informed decisions and actively participate in AI-enabled tasks. [3][6][7] User empowerment also entails considering user feedback throughout the design process, incorporating their perspectives, and ensuring that AI systems adapt to their changing needs.

Ethical considerations are integral to the development of human-centered AI. AI technologies have the potential to amplify existing biases, perpetuate discrimination, and infringe upon privacy if not designed with ethical guidelines in mind. Therefore, HCAI emphasizes the importance of fairness, transparency, accountability, and privacy protection in AI systems. Fairness ensures that AI systems do not exhibit discriminatory behavior, they consider privacy matters properly and provide equitable outcomes for all individuals [8][9][10]. Transparency enables users to understand how AI systems arrive at their decisions, promoting accountability and mitigating the "black box" problem. Privacy protection ensures that user data is handled securely and that individuals' privacy rights are respected. Effective collaboration between humans and AI is another critical aspect of HCAI. Rather than replacing humans, AI systems should augment human capabilities, enabling them to achieve tasks more efficiently and effectively. This collaborative approach promotes shared decision-making, where humans and AI work together towards common goals. [11][12] By involving humans in the decision-making process, AI systems can leverage human expertise, intuition, and creativity, leading to better outcomes across various domains such as healthcare, education, and business.[13][14][15]

1. **User Empowerment:** Highlights the significance of user empowerment in the design of AI systems. Explores Ai experiences personalization, incorporating user feedback throughout the design process and promoting user agency. By emphasizing explainable AI, the paper enables users to understand the decision-making processes of AI algorithms, fostering trust and enabling informed decision-making.

2. **Ethical Considerations:** Addressing ethical considerations is crucial in human-centered AI development.

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The paper delves into the ethical challenges associated with AI systems and proposes strategies to address them. It emphasizes the importance of fairness, transparency, accountability, sustainability [20] and privacy protection in AI design.

2. Effective Human-AI Collaboration: The paper emphasizes the collaborative nature of human-centered AI, advocating for the augmentation of human capabilities rather than their replacement. It explores principles for effective human-AI collaboration, encouraging shared decision-making and user control. [3]

II. User Empowerment in Human-Centered AI

User empowerment is a fundamental aspect of Human-Centered Artificial Intelligence (AI), aiming to AI designs prioritizing user control, agency, and understanding. Modern systems can support humans in many complex task [61]. In the context of AI, user empowerment entails tailoring AI experiences to meet the unique needs, preferences, and goals of individual users [17][18][19]. By incorporating user feedback throughout the design process and providing mechanisms for user control, AI systems empower users to actively participate in AI-enabled tasks and decision making. This section explores various dimensions of user empowerment in human-centered AI, including personalized AI experiences, the incorporation of user feedback in the design, promoting user agency, and the importance of explainable AI in fostering user understanding and trust. By emphasizing user empowerment, we can create AI technologies that enhance user satisfaction, promote user autonomy, and ultimately deliver more meaningful and impactful experiences.

A. Personalized AI Experiences

Personalization is a key aspect of user empowerment in human-centered AI. By tailoring AI experiences to meet the specific needs, preferences, and goals of individual users, we can create more engaging and impactful interactions. [4][21]Personalization involves leveraging user data, such as past interactions, preferences, and contextual information, to adapt AI systems to the unique characteristics of each user. Through machine learning algorithms and advanced data analytics techniques, AI systems can dynamically adjust their behavior, content, and recommendations to align with user expectations, and by combining multiple users and systems data, do this even in fleet level [27] too. Personalized AI experiences not only enhance user satisfaction but also improve the relevance and effectiveness of AI systems.

For instance, in e-commerce applications, personalized product recommendations based on user preferences and purchase history can significantly increase conversion rates and customer loyalty. In educational settings, adaptive platforms can provide tailored content and learning paths to suit each student's individual learning style and pace. [22] By personalizing AI experiences, we empower users by delivering customized solutions that address their specific needs and enhance their overall engagement and success. However, it is essential to strike a balance between personalization and privacy, ensuring that user data is handled securely and transparently and that users have control over the data being used to personalize their AI experiences.

B. Incorporating User Feedback in Design

In the realm of human-centered AI, incorporating user feedback in the design process is a crucial element to ensure that AI systems align with user expectations and preferences. User feedback provides valuable insights into user experiences, needs, and pain points, allowing designers and developers to iteratively improve and refine AI systems. [24][25]

Another avenue for incorporating user feedback is through user surveys, interviews, or focus groups. These methods provide a broader perspective on user preferences, expectations, and desires. By soliciting user opinions and preferences, designers gain valuable insights into user needs and can prioritize features and functionalities that align with user priorities. [25] This user-centric approach fosters user empowerment as AI systems are tailored to address the specific pain points and requirements identified through user feedback. Incorporating user feedback also entails creating channels for users to provide feedback and suggestions. This can be achieved through user-friendly interfaces that allow users to report issues, submit suggestions, or rate the performance of AI systems. By actively engaging users in the design process, designers can gain a deeper understanding of user perspectives and integrate their valuable insights into future iterations of AI systems.

Overall, incorporating user feedback in the design of AI systems is essential to ensure that the technology remains user-centered and responsive to user needs. [29][30]By actively involving users in the design process, designers can create AI systems that are more intuitive, efficient, and aligned with user expectations.

C. Promoting User Agency

Promoting user agency is a fundamental aspect of human-centered AI, aiming to empower users by granting them control, autonomy, and the ability to make informed decisions within AI systems. User agency ensures that users are not passive recipients of AI-generated outputs but active participants who can influence and shape the outcomes of AI interactions. One way to promote user agency is by providing users with meaningful choices and options within AI systems. This involves designing interfaces that allow users to customize their experiences, adjust system settings, and select from alternative recommendations or actions. [12] By offering users a range of options, AI systems cater to diverse user preferences and empower individuals to tailor the technology to their specific needs.

Another aspect of promoting user agency is transparency and explainability in AI systems. Users should have a clear understanding of how AI algorithms work, how decisions are made, and what factors influence the outcomes. Explainable AI techniques, providing understandable explanations or visualizations of AI processes, can help users comprehend the reasoning behind AI recommendations or decisions. [10][34][35]Transparent AI systems foster trust and enable users to evaluate and assess the reliability of AI outputs, ultimately allowing users to exercise their agency in accepting or challenging AI-generated results. Promoting user agency also entails giving users the ability to provide feedback, correct errors, and influence AI system behavior. By incorporating mechanisms for user input and validation, AI systems can adapt and learn from user interactions, improving their performance and aligning with user expectations. [14][37]User feedback can be solicited through understanding participation motives [42] and with interactive interfaces,

feedback loops, or user-driven customization options, enabling users to actively participate in shaping the behavior and outcomes of AI systems.

D. Explainable AI for User Understanding and Trust

Explainable AI is a critical component of human-centered AI that focuses on providing users with a clear understanding of how AI systems arrive at their decisions or recommendations. The aim is to enhance user understanding and foster trust in AI technologies by demystifying the complex inner workings of AI algorithms. One of the key challenges with AI systems, especially those based on deep learning or neural networks, is their inherent black-box nature. These models can generate highly accurate predictions or outputs, but the reasoning behind their decisions may not be readily interpretable by humans. [39][40][41]Explainable AI techniques address this challenge by providing transparent explanations or justifications for AI-generated outcomes. Explainable AI approaches can take various forms. These models provide users with clear rules or decision paths that outline how inputs are processed, and decisions are made. [13][43]By leveraging interpretable models, users can understand the logic behind AI recommendations or decisions, instilling a sense of confidence and trust in the system. Explainable AI is particularly crucial in high-stakes domains, such as healthcare or finance, where transparency and understanding of AI-generated outputs are paramount.

III. ETHICAL CONSIDERATIONS IN HUMAN-CENTERED AI

Ethical considerations are of paramount importance in the development and deployment of human-centered Artificial Intelligence (AI) systems. As AI technologies continue to advance and become more pervasive in our lives, it is crucial to ensure that these systems uphold ethical principles, respect human rights, and align with set societal values. [46][47] The following explores the ethical challenges associated with AI systems and proposes strategies to address them. By emphasizing fairness, transparency, accountability, and privacy protection, we can foster an ethical framework for human centered AI that prioritizes user well-being, avoids biases and discrimination and promotes a more equitable and trustworthy AI environment.

A. Addressing Ethical Challenges in AI Systems

AI systems present a range of ethical challenges that need to be addressed to ensure the responsible and ethical deployment of human-centered AI. One of the key challenges is the potential for biases and discrimination within AI algorithms and datasets. AI systems learn from data, and if the data used to train them is biased or reflects societal prejudices, the AI outputs can perpetuate or amplify those biases. Addressing this challenge requires careful data collection and preprocessing techniques, as well as ongoing monitoring and mitigation of biases throughout the AI system's lifecycle. Transparency and accountability are also crucial ethical considerations in AI systems. Users should have a clear understanding of how AI algorithms make decisions and the factors influencing those decisions. Transparent AI systems provide explanations, justifications, or visualizations to help users comprehend the reasoning behind AI outputs, promoting trust and enabling users to assess the reliability of AI-generated results. Additionally, accountability mechanisms should be in place to ensure that AI system developers and operators are responsible for the outcomes of their technology and there are avenues for redress in case of errors or harmful impacts. [48][49][50]

Moreover, ethical considerations also extend to the social and economic impacts of AI systems. The potential for job displacement or unequal distribution of AI benefits raises concerns about the societal consequences of AI deployment. [52][53]Efforts should be made to ensure that AI technologies contribute to a more equitable society, creating opportunities for retraining and upskilling and promoting inclusive access to AI benefits. Addressing ethical challenges in AI systems requires a multi-faceted approach involving stakeholders from diverse disciplines, including AI researchers, policymakers, ethicists, and end-users. Collaborative efforts are needed to develop ethical guidelines, regulatory frameworks, and best practices that promote the responsible and ethical use of AI technologies. By addressing ethical challenges head-on, we can ensure that human-centered AI systems are designed and deployed in a manner that respects human rights, promotes fairness, transparency, and accountability, and ultimately benefits individuals and society as a whole.

B. Fairness and Avoiding Bias in AI Design

Ensuring fairness and avoiding bias in AI design is a critical ethical consideration in human-centered AI. AI systems are susceptible to biases that can result in discriminatory outcomes, perpetuating social inequalities and reinforcing existing biases present in the data used for training. To address this challenge, it is essential to adopt fairness aware approaches throughout the AI system's development lifecycle. Fairness in AI design requires careful consideration of the data used to train AI models. Biases present in the training data, such as racial, gender, or socioeconomic biases, can be inadvertently learned and perpetuated by AI systems. [54][56]Data collection, mining [55] and preprocessing techniques should aim to mitigate and minimize biases in the training data to ensure that AI models are trained on representative and diverse datasets.

Furthermore, fairness-aware algorithms and methodologies can be employed to counteract bias in AI design. These approaches include techniques like bias-aware model evaluation, algorithmic debiasing and fairness metrics. By actively incorporating fairness considerations into the AI design process, developers can work towards mitigating biased outcomes and promoting equitable treatment. It is crucial to engage a diverse range of stakeholders and domain experts in the design and evaluation of AI systems to uncover and address biases. By involving individuals from different backgrounds, perspectives, and experiences, designers can gain valuable insights and perspectives that contribute to more equitable and unbiased AI systems. Transparency in AI decision-making is another important aspect of fairness. Users should have access to information about how AI algorithms make decisions and the factors influencing those decisions.

C. Transparency and Accountability in AI Decision-Making

Transparency and accountability are crucial principles in the design and deployment of human-centered AI systems. Transparency refers to the ability to provide clear and understandable explanations for the decisions and outputs produced by AI algorithms. Accountability, on the other hand, refers to the responsibility and answerability of AI system developers, operators, and stakeholders for the outcomes and impacts of AI technology. Transparency in AI decision-making involves making the decision process and the factors

influencing those decisions transparent to users and stakeholders. This transparency can take different forms, such as providing explanations or justifications for AI outputs, visualizing the decision-making process, or disclosing the data and algorithms used. Transparent AI systems enable users to understand how and why AI systems arrive at specific recommendations or decisions, promoting user understanding and trust.

Accountability in AI decision-making entails holding individuals and organizations responsible for the behavior and consequences of AI systems, including the design choices made during the development process, as well as accountability for the actions and impacts of the AI system in real-world scenarios [57][58][59]. Ethical guidelines and regulatory frameworks play a significant role in promoting transparency and accountability in AI decision-making. These guidelines can provide recommendations for disclosing AI system capabilities, limitations, and potential risks to users and stakeholders. Regulatory frameworks can establish requirements for transparency and accountability, ensuring that AI system developers and operators comply with ethical standards and legal obligations.

D. Privacy Protection and Data Handling

Privacy protection and responsible data handling are critical considerations in the design and deployment of human-centered AI systems. As AI relies on vast amounts of data, including personal and sensitive information, it is essential to safeguard individuals' privacy rights and ensure that data is handled in a secure and ethical manner. Privacy protection in AI systems involves adhering to privacy regulations and best practices to safeguard individual's personal information. This includes obtaining informed consent from users for data collection and usage, ensuring data security through encryption and access controls, and implementing measures to prevent unauthorized access or data breaches. Privacy policies should be transparently communicated to users, outlining how their data will be collected, used, and stored and providing them with options to control their data-sharing preferences.

By prioritizing privacy protection and responsible data handling, AI systems can engender user trust, respect individual privacy rights, and mitigate potential risks associated with data misuse or unauthorized access. Adhering to privacy regulations, implementing robust data security measures, and providing user control and transparency contribute to building a privacy centric approach to human centered AI, fostering trust and ensuring that AI technologies are developed and deployed in a manner that respects individuals' privacy and data protection.

IV. Effective Human-AI Collaboration

Effective collaboration between humans and AI systems is crucial for harnessing the full potential of artificial intelligence while ensuring that it aligns with human values and enhances human capabilities. As AI technologies become increasingly integrated into various domains, from healthcare and education to business and transportation, it is essential to establish collaborative frameworks that leverage the strengths of both humans and AI and innovate and design them properly to digital age [38]. following explores the principles, methodologies, and challenges involved in fostering effective human-AI collaboration, with aim of optimizing the synergy between human expertise and AI capabilities to achieve superior outcomes and create a positive impact on society.

A. Collaborative Approach in Human-Centered AI

A collaborative approach in human-centered AI emphasizes the active partnership and cooperation between humans and AI systems to achieve shared goals. Rather than treating AI as a replacement for human intelligence, this approach recognizes that humans and AI possess unique strengths and can augment each other's capabilities when working together. In a collaborative framework, humans and AI systems are considered equal partners, each bringing their expertise to the table. Humans contribute domain knowledge, intuition, creativity, and contextual understanding, while AI systems bring data processing capabilities, pattern recognition, and computational power. By combining these complementary strengths, collaborative AI systems aim to achieve outcomes that are superior to what either humans or AI systems could achieve alone.

One of the key aspects of the collaborative approach is the seamless integration of AI technologies into human workflows and decision-making processes. AI systems are designed to support and enhance human decision-making, providing timely and relevant insights, automating repetitive tasks, and assisting in complex analysis. Collaboration in human centered AI also entails ongoing learning and adaptation. AI systems can learn from human feedback, iteratively improve their performance, and adapt to changing circumstances. Humans, on the other hand, can learn from AI systems, gaining new insights, market analysis, expanding their knowledge base, and refining their decision-making based on AI-generated recommendations [62][63]. This iterative learning process strengthens the collaboration between humans and AI, leading to continuous improvement and innovation.

B. Shared Decision-Making and User Control

Shared decision-making and user control are fundamental principles in human-centered AI that aim to empower individuals and ensure that AI systems serve as tools to support human agency and autonomy. Shared decision-making involves involving users in the decision-making process alongside AI systems. Rather than AI making decisions on behalf of users, the focus is on creating a collaborative environment where users have the ability to influence and participate in decision-making. AI systems provide users with relevant information, insights, and recommendations, enabling them to make informed decisions based on their expertise, values, and preferences. This shared decision-making approach recognizes the importance of human judgment and respects individual autonomy.

User control is closely related to shared decision making and emphasizes the importance of putting users in control of their interactions with AI systems. Users should have the ability to understand and influence the behavior of AI systems according to their needs and preferences. This includes providing options to customize AI system settings, allowing users to define the level of automation or assistance they desire, and enabling users to provide feedback or correct AI generated outputs. Shared decision-making and user control also contribute to user trust and acceptance of AI systems. When individuals feel that they have a say in the decision-making process and have control over AI systems, they are more likely to trust and rely on AI-generated outputs.

Additionally, user control helps mitigate concerns about AI biases or errors, as users can provide feedback and correct potential issues, leading to improved AI system performance over time.

C. Leveraging Human Expertise and Creativity

In the realm of human-centered AI, the focus is not solely on the capabilities of AI systems but also on harnessing and integrating human expertise and creativity. Recognizing the unique cognitive abilities and domain knowledge of humans, this approach seeks to maximize the synergy between human intelligence and AI technologies to achieve innovative and meaningful outcomes. Human expertise plays a crucial role in shaping and guiding the development and application of AI systems. Humans possess deep knowledge in specific domains, understand complex social dynamics, and have the ability to reason, interpret, and contextualize information in ways that AI systems currently cannot. By leveraging human expertise, AI systems can benefit from nuanced insights, critical thinking, and the ability to understand the broader context in which AI is being deployed.

To effectively leverage human expertise and creativity, it is necessary to create collaborative environments that foster meaningful interactions between humans and AI systems. This involves designing AI interfaces that are intuitive and user-friendly, allowing humans to easily communicate and interact with AI technologies [64][65]. It also requires AI systems to be transparent, explainable, and adaptable, enabling humans to understand and trust the AI outputs while providing opportunities for humans to shape and refine the AI's behavior. By embracing and integrating human expertise and creativity, human centered AI systems can achieve outcomes that are more comprehensive, innovative, and aligned with human needs. The collaboration between humans and AI facilitates the exploration of new possibilities, enhances decision making processes, and ultimately contributes to the development of AI technologies that enhance human capabilities, address complex challenges, and generate positive societal impact.

D. Enhancing Productivity and User Satisfaction

A key objective of human-centered AI is to enhance productivity and user satisfaction by leveraging the capabilities of AI systems in collaboration with human users. By automating routine tasks, utilizing RPA tools integration [68], providing intelligent recommendations, and augmenting human abilities, AI technologies have the potential to significantly increase efficiency, streamline processes, and improve overall productivity. AI systems can handle repetitive, time-consuming tasks, allowing humans to focus on more complex and value-added activities. This automation of mundane tasks not only saves time but also reduces human errors and fatigue, leading to increased productivity and improved work quality. AI-powered tools, such as virtual assistants or automated data analysis systems, enable users to offload tedious tasks and allocate their time and energy to higher-level cognitive activities that require human creativity, critical thinking, and decision-making.

The collaboration between humans and AI technologies in enhancing productivity and user satisfaction is a dynamic process. Through iterative learning and feedback mechanisms, AI systems can adapt to user behaviors, preferences, and evolving needs. This ongoing interaction and improvement cycle allows AI technologies to continuously refine their performance and align with user expectations, leading to higher levels of user satisfaction. By optimizing productivity and user satisfaction, human-centered AI systems strive to create positive user experiences, improve work efficiency, and foster user trust and acceptance. The integration of AI technologies as collaborative tools that enhance human capabilities and enable users to achieve their goals more effectively contributes to a harmonious and mutually beneficial relationship between humans and AI.

V. Discussion and Conclusion

The study explored the vital aspects of human-centered artificial intelligence (AI) and its related implications, shedding light on key principles, methodologies, and challenges associated with developing AI systems that prioritize human values, user empowerment, and collaborative decision-making. Through an examination of user empowerment, ethical considerations, effective human-AI collaboration, and the importance of leveraging human expertise, we have underscored the significance of placing humans at the center of AI design and deployment. The findings demonstrate the immense potential of human-centered AI in transforming industries and improving user experiences. By incorporating user feedback in the design process, we enable users to have a sense of ownership and control over AI systems, fostering trust and acceptance. The notion of promoting user agency not only empowers individuals but also contributes to the development of AI technologies that align with human needs and values.

Ethical considerations have emerged as a critical area in human-centered AI. The analysis highlighted the importance of addressing ethical challenges in AI systems. By emphasizing fairness, avoiding bias, ensuring transparency and accountability, and protecting privacy and data handling, we can build AI systems that operate ethically and responsibly. This focus on ethical considerations paves the way for the development and deployment of AI technologies that are aligned with societal values, reduce discrimination, and mitigate potential risks. The concept of effective human-AI collaboration has emerged as a central theme throughout this research paper. By leveraging the strengths of both humans and AI, we can achieve outcomes that are superior to what either can achieve alone. Through shared decision-making and user control, we empower individuals and ensure that AI systems serve as tools to augment human intelligence and enable users to make informed decisions. The seamless integration of AI technologies into human workflows, coupled with transparent communication and iterative learning, fosters a collaborative environment where humans and AI systems work together to achieve common goals.

Moreover, the value of human expertise and creativity in the context of AI cannot be overstated. By leveraging human knowledge and creative thinking, AI systems can benefit from nuanced insights, critical reasoning, and imaginative problem-solving approaches. The collaboration between human intelligence and AI capabilities leads to innovative solutions and novel approaches that have the potential to transform industries, drive discoveries, and address complex challenges. In conclusion, this research paper has provided a comprehensive understanding of the principles, methodologies, and implications of human centered AI.

By prioritizing user empowerment, ethical considerations, effective collaboration, and the leveraging of human expertise,

we can design and deploy AI systems that enhance productivity, foster user satisfaction, and align with human values. Further, we suggest utilizing more community wisdom [51], when designing future user centric solutions.

The insights gained from this study contribute to the ongoing discourse surrounding the development of AI technologies that prioritize the needs and well-being of individuals, while driving innovation and societal progress. It is our hope that this research will inspire further exploration and advancements in the field of human centered AI, ultimately shaping a future where AI technologies truly empower and augment human capabilities.

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